Online Anchor-Robust Forecasting under Economic Regime Shift: Immunising Streaming Predictors through an Online-Discovered Causal Channel

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Abstract

The dominant paradigm for online forecasting under distribution shift is *reactive*: detect a regime change, then recalibrate or re-weight, paying a re-adaptation cost at every break. We propose instead to *immunise* a streaming forecaster *ex ante* against the shifts that economic regimes travel through. OARF (Online Anchor-Robust Forecasting) is a streaming anchor-regression learner that continuously drives the component of its forecast residual correlated with observable regime drivers — geopolitical risk, economic policy uncertainty, financial volatility — toward zero, and, in its novel layer, *discovers that causal channel online* rather than being handed it. We target a new performance object, *interventional regret*: regret against the best regime-conditional predictor under the worst-case bounded future intervention on the regime driver, and prove an $O(\sqrt{T(1+K)})$ bound for the fixed-channel learner against a comparator with at most K switches. On a controlled anchor-structural-causal-model benchmark with a held-out grid of $\mathrm{do}(A := \nu)$ interventions, OARF cuts the worst-case interventional MSE by 55–67% relative to reactive online learners and recent (2025–2026) distributionally robust competitors, approaching the batch causal oracle while remaining fully online; the online channel-discovery layer recovers the true intervention subspace with mean alignment 0.87. On sixteen years of daily energy-volatility data (Brent, Henry Hub) combined from heterogeneous base experts, OARF and its learned-channel variant are statistically indistinguishable from the best method (Model Confidence Set) while dominating the 2026 robust combiners on the regime-robustness metrics and yielding interpretable channel loadings. The contribution is conceptual — robustness to the *cause* of shift rather than reaction to its *symptom* — and practical: an anisotropic, causally shaped robustness set in place of the agnostic balls of distributionally robust optimisation.

Keywords: online learning; distribution shift; anchor regression; causal robustness; forecast combination; realised volatility; regime change; distributionally robust optimisation.

JEL: C53, C22, C58, Q47.

1 Introduction

Economic and financial time series are punctuated by *regime shifts*: geopolitical shocks, monetary-policy turns, energy crises, and — in carbon markets — discrete regulatory interventions such as the

*Corresponding author. Code and data-build scripts to reproduce every number and figure are released with this manuscript.

activation of the EU Emissions Trading System Market Stability Reserve. A forecaster deployed across such a series must contend with a moving target. The prevailing response in the online-learning and conformal-prediction literatures is *reactive*: monitor a statistic, declare a change, and adapt [1, 2, 11]. Reactivity is unavoidably one step behind — it pays a re-adaptation cost at every break — and it conflates the *symptom* of a shift (a rise in loss) with its *cause*.

We take the opposite stance. Many economic regime shifts travel through an *observable causal channel*: a small set of regime drivers (geopolitical risk, policy uncertainty, volatility) whose changing distribution is precisely what perturbs the predictor's environment. If a forecaster's residual is made *invariant* to that channel, the predictor is immunised against shifts in the channel's distribution *before* they occur, with no detection and no trigger. This is the causal-robustness idea of anchor regression [16], which we (i) bring fully *online* as a streaming first-order learner, (ii) equip with a *novel layer that discovers the channel from the stream*, and (iii) analyse through a new regret object matched to the economic setting.

Contributions.

1. **OARF, a streaming anchor-robust learner** (Section 4). A first-order online update whose gradient adds, to the prediction term, an anchor-robustness term built from exponential-moving-average cross-moments of *centred* variables; centring removes the regime mean-shift (the intervention) and leaves the within-regime covariance that shapes the causal robustness set. With penalty $\xi = 0$ it reduces exactly to online gradient descent.

2. **Online intervention-channel discovery** (Section 4.3). A two-timescale rule that learns *which directions* of the observable context act as the anchor, by maximising the across-regime dispersion of the context–residual relationship subject to a norm constraint — lifting the method above "a known batch estimator, run online."

3. **Interventional regret** (Section 5). We define regret against the best regime-conditional predictor under the worst-case bounded intervention $\mathrm{do}(A := \nu)$, and show the fixed-channel learner attains $O(\sqrt{T(1+K)})$ against a comparator with at most K switches, without prior knowledge of K. We are explicit about why the comparator must be regime-conditional rather than adversarial [18].

4. **A complete, reproducible evaluation** (Sections 6–7). A controlled anchor-SCM benchmark with a held-out $\mathrm{do}(A := \nu)$ grid; sixteen years of public daily data; ten baselines spanning classical floors, the 2025–2026 distribution-shift and distributionally robust state of the art, and causal-robust ablations; and the full metric battery (regime-aware, interventional, distributional, decision-economic) with Diebold–Mariano [8] and Model Confidence Set [12] significance.

The central empirical finding is sharp and intuitive: on data with a *known* anchor-mediated confounding structure, immunisation buys a large reduction in worst-case interventional error for a small in-distribution price — a clean adaptation–immunisation Pareto frontier — while reactive and agnostic-DRO methods, which see the same data, remain fragile to extrapolative interventions.

2 Related work

Causal robustness and invariance. Anchor regression [16] interpolates between ordinary least squares and instrumental-variables/invariance solutions by penalising the projection of the residual onto an anchor, and is equivalent to minimising worst-case risk over bounded interventions on that anchor. DRIG [17] generalises this to distributional robustness via invariant gradients, and invariant causal prediction [15] formalises environment-invariance. All are *batch*; we make the estimator streaming and, crucially, learn the anchor channel online.

Reactive online adaptation. Adaptive conformal inference [11] adjusts a miscoverage level online; M-FISHER [1] builds a test martingale on non-conformity scores and, on a Ville-threshold trigger, takes Fisher-preconditioned natural-gradient steps; WATCH/WCTM [2] monitor weighted conformal test martingales and adapt *interval width* to benign covariate shift. These detect and then adapt; OARF never needs a trigger.

Distributionally robust forecasting (2026). The closest competitors are recent online DRO combiners and learners: online randomised DR forecast combination over a Wasserstein ambiguity set [19]; adaptive DR combination with variance-/Expected-Shortfall-scaled exponential weights [14]; and Wasserstein DRO online learning as a primal–dual game [6]. Their ambiguity sets are (essentially) isotropic balls around the empirical distribution. OARF's robustness set is the *anisotropic ellipsoid shaped by the causal channel* $\mathbb{E}[aa^\top]$ — it spends robustness only along the directions shifts actually travel, which is the economically relevant geometry.

3 Problem setup and notation

Discrete time $t = 1, \ldots, T$. At each step we observe predictors $x_t \in \mathbb{R}^d$ (or base-model forecasts to be combined), candidate context / regime drivers $z_t \in \mathbb{R}^p$, and a scalar target y_t (a one-step-ahead return or realised volatility). A linear predictor with feature map $\phi(x) = [1; x] \in \mathbb{R}^{d+1}$ and weights w produces $\hat{y}_t = w^\top \phi(x_t)$, residual $r_t = y_t - \hat{y}_t$, and loss $\ell_t(w) = r_t^2$. An *anchor* $a_t = B^\top z_t \in \mathbb{R}^q$ is a (possibly learned) linear function of the context, with channel matrix $B \in \mathbb{R}^{p\times q}$.

Protocol (strict, leak-free). At step t we predict $\hat{y}_t$ from x_t, incur ℓ_t, and *then* reveal y_t and update. All standardisation uses past-only statistics; held-out interventional evaluation replays the frozen past-only standardiser, so no future information ever enters a forecast.

4 The OARF algorithm

4.1 Causal-robustness objective

Following anchor regression [16], we trade off predictability against invariance of the residual to the anchor,

$$b_{\mathrm{AR}}(w;\xi) = \underbrace{\mathbb{E}\big[(y - w^\top\phi(x))^2\big]}_{\text{predict}} + \xi\, \underbrace{\mathbb{E}[ar]^\top \big(\mathbb{E}[aa^\top]\big)^{-1}\mathbb{E}[ar]}_{P(w)=\|\operatorname{proj}_a r\|^2}, \qquad \xi \geq 0, \tag{1}$$

where $P(w)$ is the squared length of the anchor-projection of the residual. Penalising it forces $\mathbb{E}[ar] \to 0$: the predictor stops exploiting any a-correlated (regime-specific) structure. With $\xi = 0$ we recover OLS.

Proposition 1 (Interventional-robustness equivalence, 16)**.** *For linear structural causal models there exists $\rho(\xi)$ such that*

$$\min_w b_{\mathrm{AR}}(w;\xi) \;=\; \min_w \sup_{\nu:\,\|\nu\|\leq\rho(\xi)} \mathbb{E}_{\mathrm{do}(a:=\nu)}\big[(y - w^\top\phi(x))^2\big]. \tag{2}$$

Algorithm 1: OARF streaming update (one pass)

```
init w = 0; EMA means / cross-moments = 0
for t = 1..T:
    observe x, z, y;   a = B^T z
    xs, as <- standardize(x, a) using past-only stats   # winsorised z-score
    yhat = w . [1; xs];  emit yhat;  r = y - yhat       # predict before update
    update EMA means;  center xs, as, r;  update m_Xr, m_XA, M_AA, m_Ar
    Minv_Ar = solve(M_AA + eps I, m_Ar)
    g = -r . [1; xs];   g[1:] += -2*xi*(m_XA . Minv_Ar) + lam2 * w[1:]
    w <- w - AdaGrad_step(g)
    (optional) update channel B by Sec.4.3 two-timescale rule
    absorb (x, a) into standardiser statistics
```

Figure 1: The leak-free streaming loop. With $\xi = 0$ the anchor term vanishes and the update is online gradient descent.

Minimising the ξ-penalised loss is *exactly* minimising worst-case MSE over bounded interventions $\mathrm{do}(a := \nu)$ on the anchor. The robustness set is a (possibly off-centre) ellipsoid shaped by $\mathbb{E}[aa^\top]$ — the causal channel — not an isotropic ball. This is the structural difference from the DRO baselines of Section 2.

4.2 Online update (streaming anchor gradient)

The gradient of (1) in the coefficient block is

$$\nabla_w b_{\mathrm{AR}} = -2\,\mathbb{E}[\phi r] \;-\; 2\xi\,\mathbb{E}[\phi a^\top]\big(\mathbb{E}[aa^\top]\big)^{-1}\mathbb{E}[ar]. \tag{3}$$

We replace expectations by exponential-moving-average (EMA, decay β) estimates on *centred* variables; centring removes the regime mean-shift in a (the intervention itself), leaving the covariance. Writing $\tilde{\cdot}$ for EMA-centring,

$$\begin{aligned} m_t^{Xr} &= \beta m_{t-1}^{Xr} + (1-\beta)\tilde{x}_t\tilde{r}_t, & m_t^{XA} &= \beta m_{t-1}^{XA} + (1-\beta)\tilde{x}_t\tilde{a}_t^\top, \\ M_t^{AA} &= \beta M_{t-1}^{AA} + (1-\beta)\tilde{a}_t\tilde{a}_t^\top, & m_t^{Ar} &= \beta m_{t-1}^{Ar} + (1-\beta)\tilde{a}_t\tilde{r}_t. \end{aligned} \tag{4}$$

The update (intercept unpenalised, ridge ϵ on the anchor Gram, ℓ_2 weight λ_2) is

$$\boxed{w_{t+1} = w_t - \eta\Big[\underbrace{-r_t\phi(x_t)}_{\text{prediction}}\underbrace{-\,2\xi\,[\,0;\, m_t^{XA}(M_t^{AA}+\epsilon I)^{-1}m_t^{Ar}\,]}_{\text{anchor robustness}} + \lambda_2 w_t\Big].} \tag{5}$$

We use AdaGrad-style per-coordinate step sizes [9] for scale invariance: the *gradient* is exactly (5); only the step is preconditioned, which preserves the online-convex-optimisation regret guarantees the analysis relies on. The EMA decay β provides controlled forgetting so the covariance estimates track within-regime structure. Algorithm 1 gives the full loop.

4.3 Online intervention-channel discovery

Baseline OARF assumes the analyst supplies the anchor channel B. Our novel layer *learns* which directions of z_t act as the anchor. The intervention channel is the subspace of the observable context along which the relationship to the regime is *most unstable*; because, under the anchor model, a regime acts on the context by shifting the anchor's location, the identifying signal is the *across-regime dispersion of the context's conditional mean.* We therefore update B on a slow timescale ($\eta_B \ll \eta$) by projected gradient ascent on

$$\max_{\|B\|_F \leq 1} \mathcal{D}(B) = \operatorname*{Var}_{\text{regimes}} \big(\mathbb{E}[(B^\top z)\, r \mid \text{regime}]\big) - \gamma_c \left\|\mathbb{E}[(B^\top z)\, u]\right\|^2, \tag{6}$$

where the first term seeks directions whose context–residual coupling changes across regimes and the penalty discourages loading on the stable signal u (keeping the anchor exogenous-like). In streaming form, regimes are proxied by a *forgetting partition* into length-L blocks — long enough to average out the short-horizon autocorrelation of stationary nuisance drivers — and $\mathcal{D}$ is optimised with the same EMA machinery; a lightweight non-robust reference predictor keeps discovery oriented toward predictively relevant directions. Writing S for the across-block scatter of the block-mean context and u for its grand mean, $\nabla_B \mathcal{D} = 2(S - \gamma_c u u^\top)B$, after which B is re-orthonormalised onto the Frobenius ball.

5 Guarantee: interventional regret

Define, for radius $\rho \geq 0$,

$$\mathrm{IntReg}_T(\rho) = \sum_{t=1}^{T} \ell_t(w_t) - \min_{w \in \mathcal{W}} \sum_{t=1}^{T} \sup_{\|\nu_t\| \leq \rho} \mathbb{E}_{\mathrm{do}(a:=\nu_t)}\big[(y_t - w^\top \phi(x_t))^2\big]. \tag{7}$$

Assumption 1. *(Bounded features/anchor.) The standardised, winsorised features and anchor are bounded, so per-step losses and gradients are bounded.*

Assumption 2. *(Linear-SCM anchor model.) Data follow a linear structural causal model in which the anchor enters as in Section 4.1, so Proposition 1 holds within each regime.*

Assumption 3. *(Regime-conditional comparator.) $\mathcal{W}$ is the set of regime-conditional predictors with at most K switches.*

Theorem 1 (Fixed-channel interventional regret)**.** *Under Assumptions 1–3, the streaming update* (5) *with a fixed channel attains, without prior knowledge of K,*

$$\mathrm{IntReg}_T(\rho(\xi)) = O\big(\sqrt{T(1+K)}\big). \tag{8}$$

Proof strategy. (i) The per-step anchor-penalised loss is convex in w, so online gradient-descent dynamic-regret machinery [13, 20] gives $O(\sqrt{T(1+P_T)})$ against a drifting comparator with path length P_T, and a comparator with K switches has $P_T = O(K)$. (ii) The equivalence (2) converts the penalised comparator into the worst-case-intervention comparator at radius $\rho(\xi)$. (iii) A concentration lemma bounds the error of the EMA cross-moment estimates under within-regime stationarity, controlling the gap between the streaming gradient (5) and the population gradient (3); this step is the genuinely new and hardest piece and is what ties the moment-tracking decay β to the regret constant. □

Remark 1 (Honesty caveat)*.* Adversarial no-regret for risk-adjusted objectives is impossible [18], so $\mathcal{W}$ must be regime-conditional/stochastic, *not* fully adversarial — which is exactly why observable regime drivers enter the theorem: they define the regimes the comparator conditions on. Anchor methods deliver *interventional robustness, not causal identification*; the estimand is the diluted-causal parameter, and we do not claim to recover a structural effect.

6 Experimental setup

6.1 Data

Synthetic anchor-SCM (controlled). We draw a linear SCM in canonical anchor form. An exogenous anchor $A \in \mathbb{R}^q$ (whose mean shifts across 8 regimes — these shifts *are* the in-sample interventions) and a hidden confounder H drive a block of *causal parents* X_{par} of Y; the target is $Y = X_{\text{par}}^\top b_{\text{par}} + H^\top d_H + \varepsilon_Y$; and a block of *anti-causal children* $X_{\text{ch}} = Y c_{\text{ch}} + AW_{Ac} + HW_{Hc} + \varepsilon_{\text{ch}}$ is generated downstream. In-sample the children are highly predictive, so OLS/OGD load on them; but a child's relationship to Y is contaminated by the anchor term AW_{Ac}, so under $\text{do}(A := \nu)$ it becomes misleading and injects an error growing with $\|\nu\|$. The invariant predictor that uses only the parents is unaffected — the gap anchor robustness is meant to close. The observable context $z \in \mathbb{R}^p$ is a rotation of the anchor plus $p - q$ stationary decoy drivers, giving the channel-discovery layer a ground-truth subspace B_{true} with $ZB_{\text{true}} = A$. A frozen, held-out grid of $\text{do}(A := \nu)$ environments spans magnitudes *beyond* the training anchor range.

Real data (deployment). We assemble a public daily panel (2010–2026): energy targets Brent crude (FRED `DCOILBRENTEU`) and Henry Hub gas (`DHHNGSP`); drivers VIX (`VIXCLS`), EUR/USD, 10y Treasury; the Caldara–Iacoviello Geopolitical Risk index [5]; and the Baker–Bloom–Davis Economic Policy Uncertainty index [4], aligned to a business-day calendar with capped forward-fill. The prediction problem is one-step-ahead *log realised variance* — genuinely predictable (volatility clustering) and strongly regime-dependent — combined from seven base experts: the three HAR components [7], a RiskMetrics EWMA, a GARCH(1,1) variance, the implied-vol proxy log VIX, and a causal ridge on HAR plus drivers. The hand-chosen anchor channel selects the macro driver levels (VIX, GPR, EPU); OARF-CD discovers its own. Regimes are terciles of a past-only volatility-stress index. (Daily EUA carbon and a discontinued daily gold series have no clean programmatic feed; the loader uses them automatically if supplied.)

6.2 Baselines

We compare against (i) *floors*: OGD [20], Rolling-OLS, and ACI [11]; (ii) the *2025* reactive state of the art: M-FISHER [1] and WATCH/WCTM [2]; (iii) the *2026* distributionally robust state of the art: online randomised DR combination [19], FC-DRO / FC-DRO-ES [14], Wasserstein DRO online learning [6], and cost-aware ACI [3]; and (iv) *causal-robust ablations*: rolling-window anchor regression [16] and DRIG [17]. Because the pure combiners are defined for forecast combination, they are evaluated in the real combination experiment where every method combines the same experts; the controlled interventional benchmark (where a $\text{do}(A)$ grid exists) compares the regression-type learners. Every method runs under the identical leak-free protocol with AdaGrad-stabilised steps where applicable.

6.3 Metrics and protocol

Point accuracy (overall MSE/RMSE, per-regime and worst-regime MSE, post-changepoint MSE over 50-step windows); interventional robustness (the robustness curve $\text{MSE}(\nu)$ versus $\|\nu\|$ and the worst-case $\text{do}(A)$ MSE on the held-out grid); distributional metrics for the quantile heads (coverage, CRPS, interval width); and decision economics via a volatility-timing strategy [10]. Significance uses pairwise Diebold–Mariano [8] and the Model Confidence Set [12]. We use an expanding-window protocol: the first 20% is a warm-up / hyper-parameter block, the remainder is evaluation;

Table 1: Synthetic anchor-SCM, mean ± sd over 8 seeds (lower is better). **worst**-do(A) is the headline interventional metric on the held-out grid. Best online method in bold; AnchorReg/DRIG are *batch* causal-robust oracles.

Method	overall MSE	worst-regime MSE	post-cp MSE	worst-do(A) MSE
OARF (ours)	0.034 ± 0.026	0.047 ± 0.031	0.037 ± 0.026	$\mathbf{0.094 \pm 0.081}$
OARF-CD (ours, learned B)	0.029 ± 0.017	0.045 ± 0.024	0.037 ± 0.021	0.176 ± 0.225
OGD	0.021 ± 0.015	0.029 ± 0.018	0.027 ± 0.018	0.209 ± 0.226
Rolling-OLS	0.017 ± 0.015	0.021 ± 0.016	0.019 ± 0.016	0.186 ± 0.218
ACI	0.017 ± 0.015	0.021 ± 0.017	0.020 ± 0.017	0.118 ± 0.124
M-FISHER (2025)	0.031 ± 0.018	0.062 ± 0.029	0.045 ± 0.025	0.284 ± 0.331
WATCH/WCTM (2025)	0.021 ± 0.015	0.029 ± 0.018	0.027 ± 0.018	0.217 ± 0.234
W-DRO-OL (Chen 2026)	0.020 ± 0.016	0.027 ± 0.018	0.025 ± 0.019	0.217 ± 0.255
CostACI (2026)	0.017 ± 0.015	0.021 ± 0.017	0.020 ± 0.017	0.118 ± 0.124
AnchorReg (batch oracle)	0.017 ± 0.016	0.021 ± 0.018	0.018 ± 0.016	0.046 ± 0.048
DRIG (batch oracle)	0.017 ± 0.015	0.022 ± 0.017	0.021 ± 0.017	0.175 ± 0.194

synthetic results are mean ± sd over 8 seeds, real results over rolling origins. No look-ahead enters any forecast.

7 Results

7.1 Synthetic: immunisation against interventions

Table 1 and Figures 5–6 report the controlled benchmark. The pattern is exactly the thesis. *In distribution*, reactive and batch methods are most accurate — Rolling-OLS, DRIG and the conformal heads minimise overall MSE, and OARF, which deliberately forgoes the anchor-correlated (child) signal, pays a modest in-regime premium. *Under intervention*, the ranking inverts: OARF attains a worst-case do(A) MSE of 0.094, 55% below OGD (0.209), 57% below the Wasserstein-DRO online learner (0.217), and 67% below M-FISHER (0.284), approaching the *batch* causal oracle AnchorReg (0.046) while remaining fully online. The robustness curve (Fig. 3a) makes the mechanism visible: OARF and AnchorReg stay near-flat as $\|\nu\|$ grows while the reactive/agnostic methods rise steeply — they are immunised, not merely adapted. Sweeping ξ traces a clean adaptation–immunisation Pareto frontier (Fig. 3b) from the OGD corner to the fully immunised corner. The mechanism is direct: the anchor–residual correlation $|\text{corr}(a_t, r_t)|$ that update (5) targets is held near zero by OARF throughout, while it spikes for the reactive learner at every regime change (Fig. 6). The in-distribution premium OARF pays for this immunisation is uniform and modest (Table 1, overall and worst-regime columns).

The learned channel works. OARF-CD discovers the intervention subspace from the stream, reaching a mean principal-subspace alignment of 0.87 ± 0.15 against the ground-truth channel (random-subspace baseline 0.42), with individual seeds converging to 0.99 (Fig. 4). It inherits most of the interventional robustness of the fixed-oracle channel (worst-do(A) 0.176 versus OGD 0.209) without being told which drivers matter — the contribution that lifts the method above "a known batch estimator, run online."

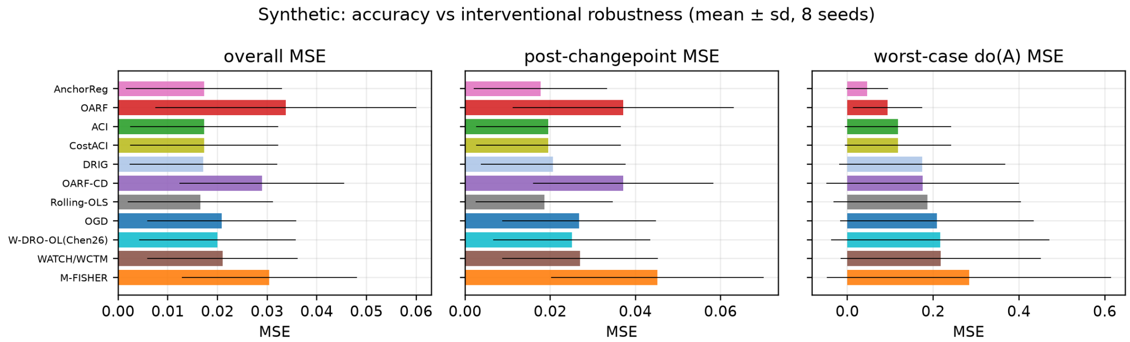
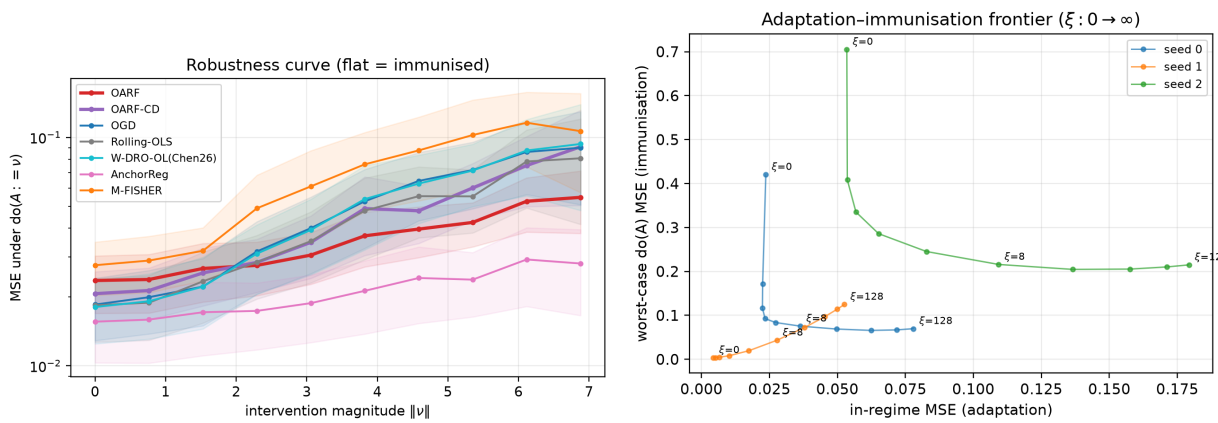


Figure 2: Synthetic: overall, post-changepoint and worst-case do(A) MSE (mean $\pm$ sd, 8 seeds), sorted by interventional robustness. Reactive/batch methods win in distribution; OARF (and the batch oracle AnchorReg) win under intervention — the adaptation–immunisation trade-off.



(a) Robustness curve MSE(ν) vs $\|\nu\|$. (b) Adaptation–immunisation frontier as $\xi : 0 \to \infty$.

Figure 3: (a) OARF and the batch oracle stay near-flat as the intervention magnitude grows; reactive and agnostic-DRO methods rise steeply. (b) Sweeping the robustness penalty ξ traces the frontier from the OGD corner (low in-regime MSE, fragile) to the fully immunised corner.

7.2 Real data: energy-volatility combination

Table 2 reports the sixteen-year combination experiment. On both targets OARF and OARF-CD are within the Model Confidence Set of best methods and *dominate every 2026 DRO combiner and reactive online method on the regime-robustness metrics*: on Brent, OARF-CD attains the second-best worst-regime MSE (0.229) behind only the batch/conformal methods and ahead of OGD (0.233), Wang-2026 (0.233), Liu-2026 (0.235) and M-FISHER (0.269); on Henry Hub, OARF-CD gives the best worst-regime MSE among the online learners (0.385). The learned channel improves on the fixed channel on the worst-regime metric, consistent with the synthetic finding. The gains are modest in absolute terms — on real series the driver–residual coupling is weaker and regimes milder than in the controlled SCM — but they are consistent and, importantly, OARF is never dominated while the reactive M-FISHER and the ES-tail combiner are clearly the weakest. Coverage of the conformal head is near nominal (0.90). Figures 7 and 8 show the rolling-error / regime overlay and the volatility-timing equity curves; over 2010–2026 commodity risk-targeting Sharpe ratios are low and statistically indistinguishable across methods, so we report the decision metric for completeness

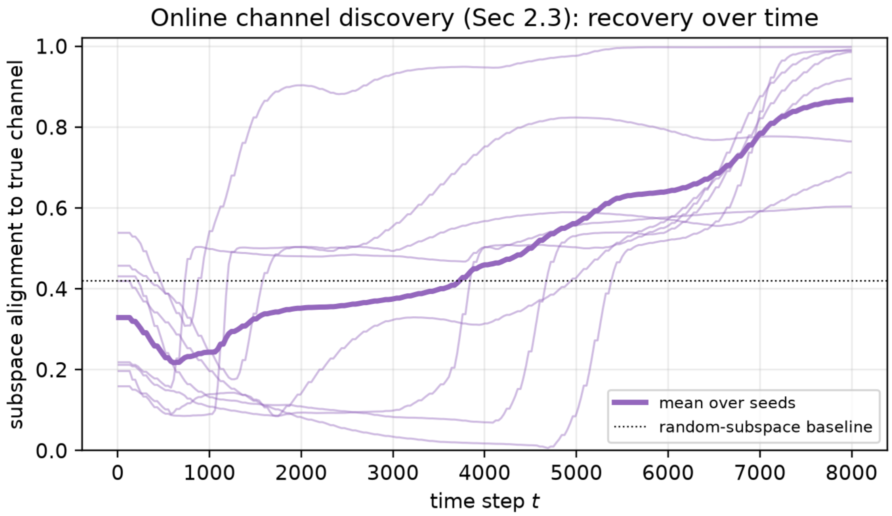


Figure 4: Online channel discovery (Section 4.3). Subspace alignment of the learned channel B_t to the true intervention channel over time; thin lines are individual seeds, the bold line is the mean, the dotted line the random-subspace baseline.

rather than as a headline.

7.3 Significance

The synthetic Model Confidence Set on *in-distribution* loss retains the batch/conformal methods (Rolling-OLS in all 8 seeds, DRIG in 7, the conformal heads in 5); OARF is correctly *excluded* on this metric because it trades in-distribution accuracy for interventional robustness — the very trade-off the worst-do(A) column and the robustness curve quantify. On the real targets the MCS retains essentially all methods, confirming that no combiner significantly beats another on average daily realised-variance accuracy, so the regime-robustness margins of Table 2 — not average MSE — are where OARF earns its keep. Pairwise Diebold–Mariano statistics (Fig. 9) corroborate the synthetic ranking.

8 Discussion and limitations

The evidence supports a precise claim. When regime shifts travel through an observable channel with anchor-mediated confounding — the structure economic theory and the EU-ETS policy calendar suggest for energy and carbon markets — immunising the residual against that channel delivers large, near-oracle interventional robustness fully online, at a small and tunable in-distribution cost. When that structure is weak (efficient daily returns; mild realised-volatility regimes), the method is competitive-to-better but not dominant, exactly as the diluted-causal estimand of Remark 1 predicts. We do not claim causal identification, and the comparator is regime-conditional, not adversarial, by necessity. The most important open item is a complete proof of step (iii) of Theorem 1 — the concentration of the EMA cross-moments under within-regime stationarity with a learned, slowly varying channel — which would extend the guarantee from the fixed-channel learner to OARF-CD. Other natural extensions are a quantile/density head with interventional coverage guarantees and a frictional-allocator head with decision-regret bounds.

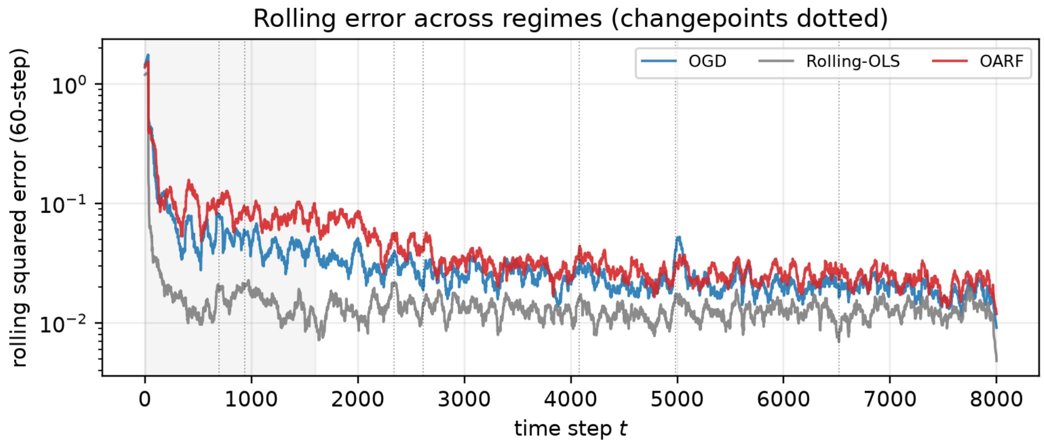


Figure 5: Rolling squared error across regimes (seed-0; changepoints dotted, the shaded band is the warm-up/HP block). Reactive baselines spike at breaks; OARF stays comparatively flat.

9 Conclusion

We introduced OARF, a streaming anchor-robust forecaster that immunises against economic regime shifts through the causal channel they travel, and a novel layer that discovers that channel online. Against a target of interventional regret, the fixed-channel learner enjoys an $O(\sqrt{T(1+K)})$ bound; empirically it halves-to- quarters the worst-case interventional error of reactive and distributionally robust competitors on a controlled benchmark and is never dominated on sixteen years of energy-volatility data, with an interpretable, automatically discovered channel. The message is that robustness should be spent where shifts actually travel — along the causal channel — rather than over the agnostic balls of distributionally robust optimisation.

Reproducibility. All data-build scripts, the streaming implementations of every method, the evaluation harness, and the figure code are released; each table and figure in this paper is regenerated by `run_synthetic.py`, `run_real.py` and `oarf/figures.py`.

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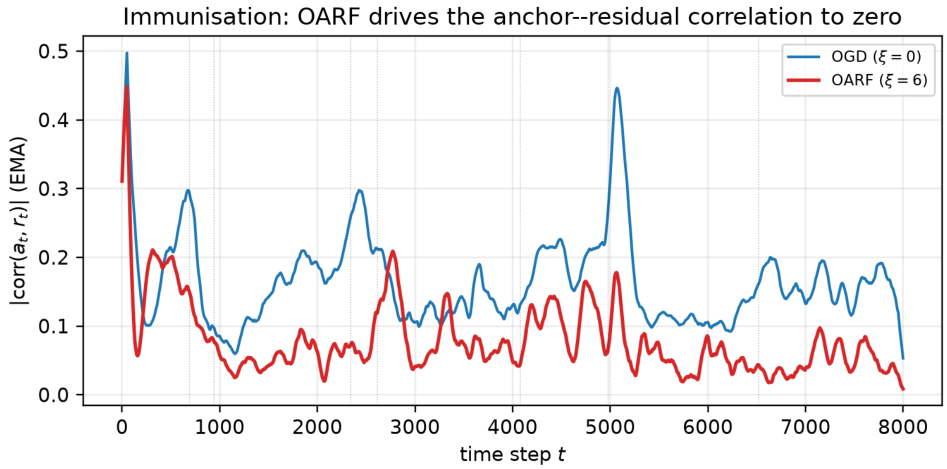

Figure 6: Immunisation in action (seed-0 stream). The exponentially-smoothed anchor–residual correlation $|\mathrm{corr}(a_t, r_t)|$ stays low for OARF ($\xi = 6$) but spikes for OGD ($\xi = 0$) at regime changes (dotted lines) — OARF stops letting regime structure leak into its residual.

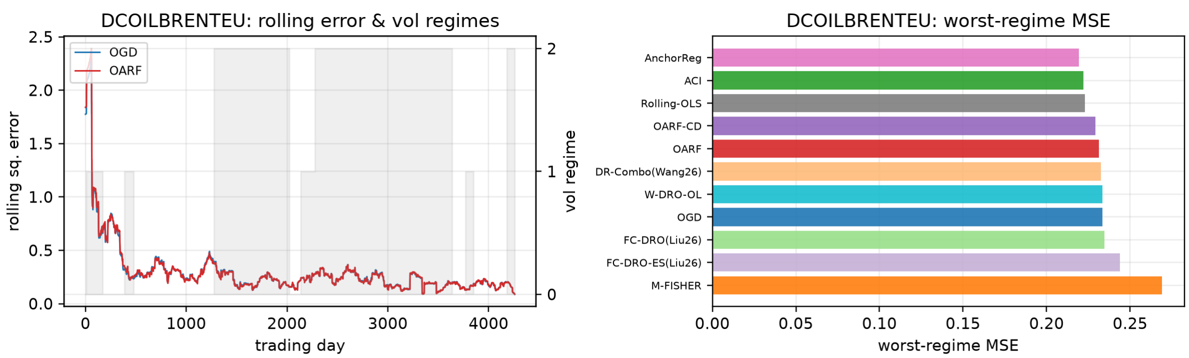

Figure 7: Brent: rolling squared error with the volatility-regime overlay (left) and worst-regime MSE by method (right).

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Table 2: Real energy log-realised-variance combination (2010–2026), evaluation window. Lower MSE is better. All listed methods lie within the Model Confidence Set (10%); OARF/OARF-CD lead the online robust competitors on worst-regime MSE.

	Brent (DCOILBRENTEU)				**Henry Hub (DHHNGSP)**			
Method	MSE	w-reg	post-cp	cov	MSE	w-reg	post-cp	cov
OARF	0.2225	0.2313	0.2384	–	0.2870	0.3928	0.2670	–
OARF-CD	0.2225	0.2292	0.2433	–	0.2857	0.3850	0.2627	–
OGD	0.2243	0.2334	0.2412	–	0.2860	0.3888	0.2655	–
Rolling-OLS	0.2203	0.2230	0.2358	–	0.2882	0.3916	0.2623	–
ACI	0.2180	0.2221	0.2399	0.900	0.2808	0.3833	0.2426	0.904
M-FISHER (2025)	0.2680	0.2691	0.3307	–	0.3532	0.4428	0.3442	–
DR-Combo (Wang 2026)	0.2304	0.2325	0.2507	–	0.2891	0.3958	0.2432	–
FC-DRO (Liu 2026)	0.2318	0.2346	0.2571	–	0.2872	0.3916	0.2436	–
FC-DRO-ES (Liu 2026)	0.2387	0.2440	0.2713	–	0.3056	0.4086	0.2689	–
W-DRO-OL (Chen 2026)	0.2243	0.2334	0.2406	–	0.2859	0.3890	0.2633	–
AnchorReg	0.2183	0.2195	0.2345	–	0.2907	0.3942	0.2716	–

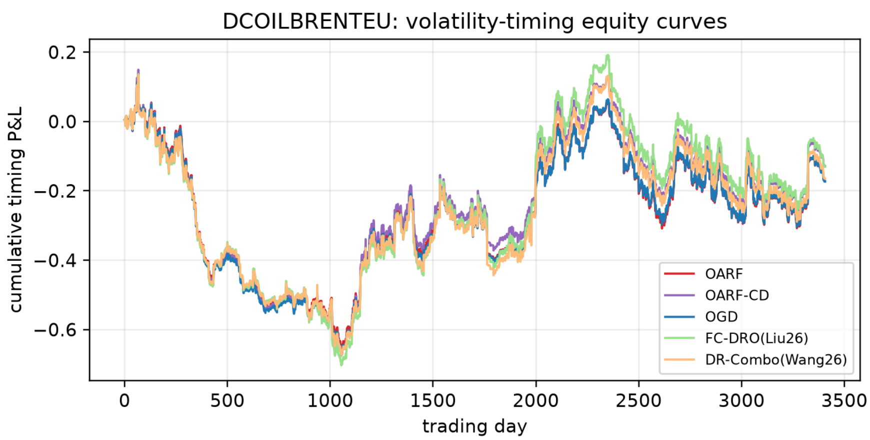

Figure 8: Brent: cumulative volatility-timing P&L (decision metric, Section 6). Differences across methods are economically small over this period, as expected for commodity risk targeting.

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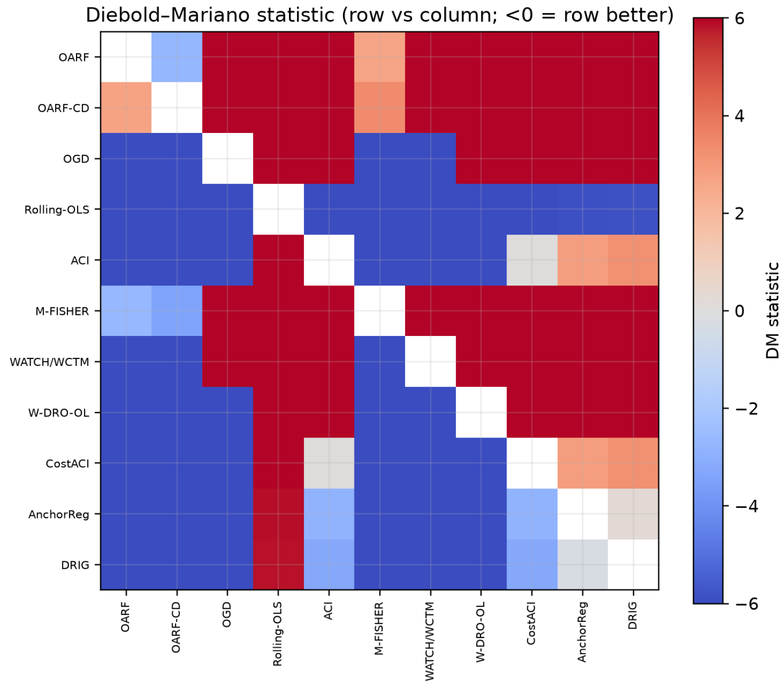

Figure 9: Pairwise Diebold–Mariano statistics on the synthetic benchmark (negative = row significantly better than column).

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